CREATING JOBS AND INTEGRATING GIT IN JENKINS

CREATING 1ST JOB

1. name ------> my-1st-job
2. freestyle project
3. description ---------> These is my 1st job
4. add build steps ----> execute shell
    1. echo “Hello World”
5. apply and save
6. build now

PULL THE PUBLIC REPOSITORY

1. create a job
    1. name ----> git-pull-job
    2. freestyle project
    3. this project is parameterized ------> string parameter
       name ----> name
       value-----> cloudblitz
    4. build steps
    5. execute shell
        a. git clone (paste repo)
    6. apply and save

2. build now
3. check the console output

OR

1. create a job
    1. name --------git-pull-job
    2. freestyle project
    3. This project is parameterized------string parameter
       name ----> name
       value-----> cloudblitz
    4. scm-------> git-------> {https://github.com/mayurpandit25/git.git}
    5. apply and save

2. build now
3. check the console output

HOW TO ADD THE CREDENTIALS FOR GITHUB ---> HTTPS

1. go to github and create the access token (classic)
    1. setting
    2. developer setting
    3. PAT token (classic)
    4. generate new token
    5. copy these token

1. go to manage Jenkins
2. click on global credentials
3. add the credentials
        1. kind ----------> username and password
        2. username -----> mayurpandit25
        3. password -----> (paste the token here)
        4. create the credentials

PULL THE PRIVATE REPOSITORY

1.create a job
    1.name -----> gitpull
    2.scm--------> git
        1. repository url------> (repo url)
        2. credentials ---------> (add credentials u created already)
        3. branch name ------> main
2. build now
3. check the console output

EXPLANATIONS

1. Discard Old Builds

Automatically delete old build history to save disk space.

Practical

1. Name -------> my-2nd-job
2. freestyle project
3. Discard old builds
    1. log rotation
    2. Days to keep builds --------> 7
    3. Max # of builds to keep ---> 10
4. execute shell
    1. echo “hello”

5. apply & save
6. build now

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2. This Project is Parameterized

Allow users to provide input values before running a build.

Practical

1.name ---------> my-3rd-job
2.freestyle job
3. This project is parameterized --> choice parameter
   1. Name: ENV
   2.Values:
      1. dev
      2. test
      3. prod
4.execute shell
  echo "Deploying to $ENV"
5.apply & save
6.build with parameters (give any parameter)
  Deploying to test

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3. Throttle Builds

Limit the number of builds running simultaneously.

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4. Execute Concurrent Builds if Necessary

Allow multiple executions of the same job at the same time.

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5. Restrict Where This Project Can Be Run

Run a job only on specific Jenkins agents (workers).

TRIGGERS

1. Build after other projects are built

Run this job only after another Jenkins job completes.

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2. Build periodically

Run a job automatically at a fixed schedule regardless of code changes.

Every minute: * * * * *

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3. GitHub hook trigger for GITScm polling

Build immediately whenever code is pushed to GitHub.

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4. Poll SCM

1. Jenkins periodically checks Git for changes.
2. If changes are found, Jenkins starts a build.
3. Every minute: * * * * *
